

# Tao Jing (荆韬)

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## Research Interests

- Investigate stellar feedback and its impact on star formation and quenching processes at sub-galactic scales using multi-band imaging and integral field spectroscopy.
- Study the interactions within the multiphase interstellar medium.
- Explore the microstructure of shock- or photon-ionized nebulae in the Milky Way and nearby galaxies to refine ionization models and improve hydrodynamic simulations.
- Develop and tailor advanced machine learning techniques for efficient analysis and interpretation of astronomical datasets.

## Education

- **Doctor of Philosophy**, Astronomy  
Department of Astronomy, Tsinghua University (THU) 2020 - Now
- **Bachelor of Science**, Astronomy  
Department of Astronomy, Xiamen University (XMU) 2016 - 2020

## Publications

- **Jing, T.** and Li, C. (2025), Regression for Astronomical Data with Realistic Distributions, Errors, and Non-linearity. *AJ*, 170, 45, 10.3847/1538-3881/add891
- **Jing, T.** and Li, C. (2024), On the Origin of Quenched but Gas-rich Regions at Kiloparsec Scales in Nearby Galaxies. *ApJ*, 975, 17, 10.3847/1538-4357/ad7367
- **Jing, T.**, Li, C., Yan, R., Cheng, C., Zhang, W., Ji, X., Li, N., Wang, J., Wu, C., and Yuan, H. (2024), Criss-cross Nebula: A Case Study of Shock Regions with Resolved Microstructures at Scales of 1000 au. *ApJ*, 962, 51, 10.3847/1538-4357/ad11d1
- Ma, C., Sun, Z., **Jing, T.**, Cai, Z., Ting, Y.-S., Huang, S., and Li, M. (2025), Can AI Dream of Unseen Galaxies? Conditional Diffusion Model for Galaxy Morphology Augmentation. *arXiv e-prints*, arXiv:2506.16233, 10.48550/arXiv.2506.16233
- Guo, R., Li, C., Zhou, S., Li, N., **Jing, T.**, and Cheng, Z. (2025), Mapping Dust Attenuation at Kiloparsec Scales. II. Attenuation Curves from Near-ultraviolet to Near-infrared. *Research in Astronomy and Astrophysics*, 25, 065017, 10.1088/1674-4527/add673
- Sui, C., Zhao, X., **Jing, T.**, and Mao, Y. (2023), Evaluating Summary Statistics with Mutual Information for Cosmological Inference. *Machine Learning for Astrophysics*, 30, 10.48550/arXiv.2307.04994

## Collaborations

- [Affordable Multiple Aperture Spectroscopy Explorer \(AMASE\) project](#) [🔗](#): a in-processing project that will pair 100 identical multi-fiber spectrographs to map the ionized gas in the Milky Way and nearby galaxies at the spatial resolution of half an arcminute and a spectral resolution of  $R=15,000$ .

I am in Science Team of AMASE and work for the HII region sample selection and construct mock data cubes from simulations. I am also leading a subproject to study microstructures in shocked regions.

## Approved Proposals

- PMO Delingha Millimeter Telescope 2022 (30.5 hours awarded) PI  
Title: Detecting  $^{12}\text{CO}$  in the Criss-cross Nebula to study shock-molecular cloud collision.
- SITELE@CFHT 2022B (5.5 hours awarded) PI  
Title: High resolution IFS on Criss-cross Nebula as a case study of microstructure of shock regions

- MeerKAT 2022 (7.5 hours awarded) PI  
Title: High resolution HI observation on Criss-cross Nebula: the case study for shock-cloud interaction
- SITELE@CFHT 2024B (PI: Xiao Li, 5.6 hours awarded) CoI  
Title: Large FoV IFS on UGC1382 as a case study of quenched but HI-rich galaxies

## Technical Skills

- **Programming Language:** Python, C, FORTRAN, JavaScript, Mathematica,  $\LaTeX$
- **Ionization Model:** [MAPPINGS V](#) [↗](#)
- **Spectrum Analysis:** [pPXF](#) [↗](#), self-developed code
- **Simulation Post-processing:** [COLT](#) [↗](#), [CHIMES](#) [↗](#)
- **Machine Learning Packages:** [TensorFlow](#) [↗](#), [JAX](#) [↗](#), [PyTorch](#) [↗](#), [scikit-learn](#) [↗](#), [AutoGluon](#) [↗](#)

## Selected Talks and Posters

- |                                                                                                                                  |      |
|----------------------------------------------------------------------------------------------------------------------------------|------|
| ○ The 12th Sino-German Meeting on Galaxy Formation<br>Poster + Flash talk, Chengdu, China                                        | 2025 |
| ○ Decoding Galactic Evolution through the Interplay of the Multi-Phase Interstellar Medium<br>Poster + Flash talk, Nagoya, Japan | 2025 |
| ○ The 12th Symposium on Interstellar Physics and Chemistry<br>Oral talk, Zhangye, China                                          | 2025 |
| ○ Symposium on Evolutionary Laws of Galactic Ecosystems over Billions of Years<br>Poster + Flash talk, Beijing, China            | 2025 |
| ○ The 27th Guoshoujing Symposium on Galaxies and Cosmology<br>Oral talk, Changsha, China                                         | 2025 |
| ○ AMASE Science Meeting<br>Oral talk, Xiangtan, China                                                                            | 2025 |
| ○ JWST Discussion Workshop<br>Invited talk, Given remotely                                                                       | 2025 |
| ○ The Ecosystem of Gas and Dust in Galaxies Near and Far<br>Oral talk, Chiang Mai, Thailand                                      | 2024 |
| ○ FAST Deep Sky Survey Annual Meeting 2024<br>Oral talk, Beijing, China                                                          | 2024 |
| ○ THU Galaxy & Cosmology Seminar<br>Invited talk, Beijing, China                                                                 | 2024 |
| ○ The 11th Symposium on Interstellar Physics and Chemistry<br>Oral talk, Nanjing, China                                          | 2024 |
| ○ AMASE Science Meeting<br>Two oral talks, Beijing, China                                                                        | 2024 |
| ○ The 26th Guoshoujing Symposium on Galaxies and Cosmology<br>Oral talk and Poster, Suzhou, China                                | 2024 |
| ○ Resolving Galaxy Ecosystems Across All Scales 2023<br>Oral talk, Hongkong, China                                               | 2023 |
| ○ ACAMAR Gas in Galaxies Workshop 2023<br>Oral talk and Poster, Perth, Australia                                                 | 2023 |
| ○ LAMOST MRS-N and Star Formation Evolution Seminar<br>Oral talk, Xuyi, China                                                    | 2023 |

- SDSS Collaboration Meeting 2021 2021  
Oral talk, Given remotely
- The 23th Guoshoujing Symposium on Galaxies and Cosmology 2021  
Oral talk, Hangzhou, China

## Teaching, Professional Services, and Student Supervision

- Qianyi Liu (刘谦益), undergrad at THU 2025 - now
- Organizing Committee of THU Data Science Club/Machine Learning Session 2021 - 2023
- Teaching Assistant, Observational Cosmology: From the Solar System to the Depths of the Universe 2020 - 2021

## Honors and Awards

- Tsinghua University Comprehensive Excellence Scholarship (First Prize) 2024 - 2025
- Tsinghua University Comprehensive Excellence Scholarship (Second Prize) 2023 - 2024
- AMD Scholarship 2020 - 2021
- Shanghai Astronomical Observatory Guangqi Scholarship (First Prize) 2018 - 2019
- Xiamen University Tan Chong International Limited Scholarship 2017 - 2018

## References

- Prof. Cheng Li Tsinghua University  
cheng.leech@gmail.com
- Prof. Jing Wang Peking University  
jwang\_astro@pku.edu.cn
- Prof. Renbin Yan The Chinese University of Hong Kong  
rbyan@cuhk.edu.hk